

RENANCE

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

CHM 213 — Past Questions & Materials
by Renance

Course: CHM 213
Institution: Federal University of Petroleum Resources, Effurun
Papers in this pack: 1
Levels: 200

This study pack was compiled and restructured by Renance for CHM 213 students at Federal University of Petroleum Resources, Effurun. It brings together 1 past paper(s)/material(s) into one clean, readable document so you can revise without chasing scattered PDFs. Practise these same questions interactively on Renance with instant answers, AI explanations, and full CBT mock exams — over 235,000+ questions across JAMB, Post-UTME, WAEC, NECO and university courses.

Study online: renance.app
Get the app: Google Play — [dev.renance.renance](https://play.google.com/store/apps/details?id=dev.renance.renance)

Generated by Renance AI

Paper 1 of 1 — CHM 213 PHYSICAL CHEMISTRY PQ 2022/23

Level 200 • First Semester • PastQuestion

FEDERAL UNIVERSITY OF PETROLEUM RESOURCES, EFFURUN

COLLEGE OF SCIENCE

DEPARTMENT OF CHEMISTRY

2022/2023 FIRST SEMESTER PRACTICE QUESTIONS

CHM 213: PHYSICAL CHEMISTRY (2 credits)

General Information:

Atomic masses: O=16, H=1, C=12, N=14, Cl= 35.5, He =4, Ar=40, Ne=20; At STP: Pressure = 1 atm = 760 torr = 760 mmHg = 1.01325 bar = 101.325 kPa = 101325 Pa., Temperature ~ One 273 K; 1 mole of gas occupies 22.4 L; $N = 6.023 \times 10^{23}$ atoms. $R = 8.314 \text{ J mol}^{-1} \text{ K}^{-1} = 0.0821 \text{ L atm mol}^{-1} \text{ K}^{-1} = 1.987 \text{ cal deg}^{-1} \text{ mol}^{-1} = 0.08314 \text{ L bar mol}^{-1} \text{ K}^{-1} = 62.39 \text{ mmHg L mol}^{-1} \text{ K}^{-1}$;

1. (a) i. A person with a fever has a temperature of 101.0°F, What is this temperature in degrees Kelvin?

(ii) A sample of metal that has a mass of 183.7 g is added to a graduated cylinder filled with 231.0 mL of water. The level of the water rises to 240.5 mL. What is the density of the metal?

(iii) What is the density (in g/L) of a gas with a molar mass of 60 g/mol at 0.75 atm and 27 °C?

(b) i. If 4g of a gas occupies 11.2 litres at 0°C and 0.25 atm, what is the molecular mass of the gas? :

(ii) Consider these identical flasks filled with three different gases:

Flask A | CO at 760 torr and 0.0°C _|

Flask B Np at 250 torr and 0.0°C

Flask C Hy at 760 torr and 0.0°C

In which flask will the molecules have the greatest average kinetic energy?

(c) In a lecture room, the composition of air was given to be by volume 78% N₂, 21% O₂ and 1% other. If the total pressure of air inside a sealed container is 2.0 atm, what is the partial pressure of O₂?

2. Calculat

2.59×10^3 mol occupies 121mL

pressure of 1150 torr; (iii) the volume of the gas in litres
a temperature of 47°C.

3a) State the fundamental postulates on whi

b) (i) What two properties of gas molecule

these properties account for the behaviour of real gases expresse
equation.

(ii) Calculate the pressure exerted by 1 mole of water vapour in 20 litres container at 27°C

using the ideal gas law and van der Waals equation [The values of van der Waals constants a and b for water vapour are 5.464 litre² atm.mol⁻² and 0.03049 litre mol⁻¹ respectively].

c) If 12.0g of NO , 0.40g of H_2 and 9.0g of O_2 are put into a 1.00 litre container at 27°C , calculate the partial pressures of each gas and the total pressure in the container.

4. (a) Define the term "partial pressure" and state Dalton's law of partial pressures. :

(b) A mixture containing 0.432 mol of helium gas, 0.357 mol of neon gas and 0.103 mol of argon gas is confined in a 7.0 L vessel at OSE,

(c) Calculate the partial pressure of each of the gases in the mixture

(ii) Calculate the total pressure of the mixture

(c) Calculate

(i) the density of NO_2 at 1.00 atm and 25°C .

(ii) __ the molar mass of a gas if 4.40g occupies 3.50 L at 560 torr and 41°C .

5. To synthesize ammonia using the Haber process, 2000 moles of nitrogen are heated in an 800-L vessel to 625°C . Calculate the pressure of the gas if nitrogen behaves

each of the following quantities for an ideal gas: (i) the pressure in atmospheres, if

at -15°C ; (ii) the quantity of gas in moles if 4.65L at 35°C has a

if 1.76 mol has a pressure of 0.76 atm at

ch the kinetic theory is based.

s cause them to behave non-ideally? Explain how

d by the Van der Waals